7.3 Minimisation Methods

Candidate number: *insert candidate number*

insert date of final run

Computational Setup

In 3–5 sentences, state the language/environment, double precision, how λ^* was selected or checked, and how iteration data were recorded. Mention any convergence checks only if they affected the reported runs. Avoid implementation detail unless it explains a numerical result.

Table 1: Summary of reported runs.

Run	Function	Method	Starting point	Iterations reported
Q2	(4)	Steepest descents	$(\cdots, \cdots)$	$\cdots$
Q3	(5)	Steepest descents	$(\cdots, \cdots)$	$\cdots$
Q4	(4)	Conjugate gradients	$(\cdots, \cdots)$	$\cdots$
Q5	(5)	Conjugate gradients	$(\cdots, \cdots)$	$\cdots$
Q6	(6)	DFP	$(\cdots, \cdots, \cdots)$	$\cdots$
Q7	(4)	DFP	$(\cdots, \cdots)$	$\cdots$
Q8	(5)	DFP	$(\cdots, \cdots)$	$\cdots$

Question 1: Shape and Analytic Minima

Table 2: Analytic minima of the two two-variable objectives.

Function	Minimiser (x^*, y^*)	f^*
(4)	$(-0.3700, -0.7937)$	-1.0953
(5)	$(1, 1)$	0

Function (4).

- Solve $\partial_x f = 1 + x - y^2 = 0$: $x = y^2 - 1$.
- Sub into $\partial_y f = 1 - 2y + 4y(y^2 - x/2) = 0$: $1 + 2y^3 = 0 \Rightarrow y^* = -2^{-1/3}$, $x^* = 2^{-2/3} - 1$ (unique real root).
- Hessian: $H = \begin{pmatrix} 1 & -2y^* \\ -2y^* & -2+12y^{*2}-2x^* \end{pmatrix} \approx \begin{pmatrix} 1 & 1.587 \\ 1.587 & 6.300 \end{pmatrix}$; $\det H \approx 3.78 > 0$, $\operatorname{tr} H > 0 \Rightarrow$ local min.
- $(y^2 - x/2)^2 \to \infty$ as $|(x, y)| \to \infty \Rightarrow$ also global min.
- Eliminate $x = y^2 - 1$ in f: $f^* = -\frac{1}{2} - 2^{-1/3} + 2^{-7/3} \approx -1.0953$.

Function (5).

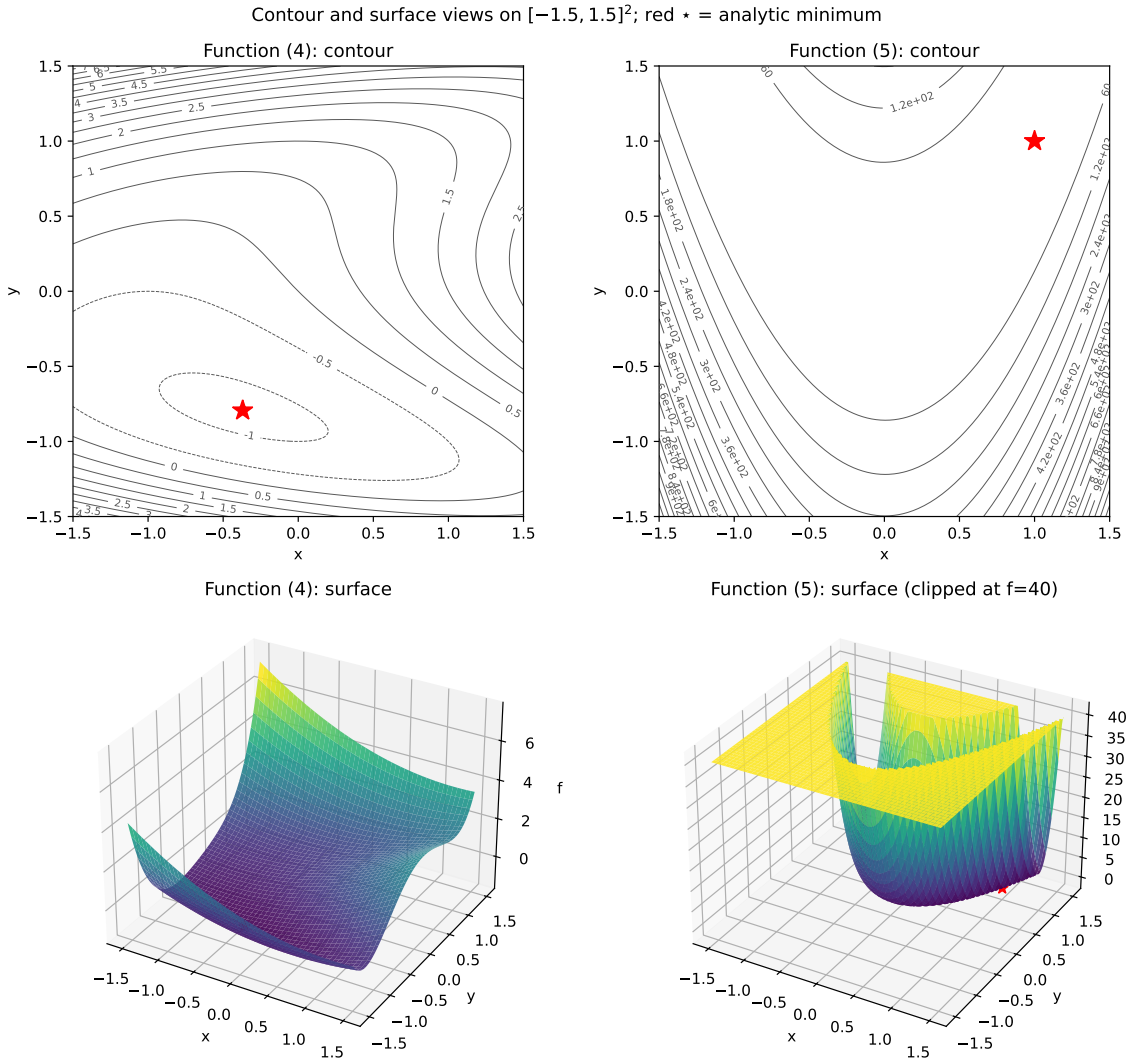

Figure 1: Contour and surface plots of functions (4) and (5) on $[-1.5, 1.5]^2$; red stars mark the analytic minima.

- $f = (1-x)^2 + 80(y-x^2)^2 \geq 0$, with equality iff $y = x^2$ and $x = 1 \Rightarrow (x^*, y^*) = (1, 1),\ f^* = 0$, unique global min.
- Hessian at $(1, 1)$: $H = \left(\begin{smallmatrix} 642 & -320 \\ -320 & 160 \end{smallmatrix}\right)$; $\det H = 320 > 0,\ \operatorname{tr} H > 0 \Rightarrow$ strict min confirmed.

Questions 2–3: Steepest Descents

Question 2: Function (4)

What to report.

- Required evidence: a path plot for the 10 steepest-descents iterations, plus a trace table containing λ^*, $f(x_n)$, and the decrease after each step.
- From the numerical run alone, give an estimated limiting minimum value and intervals for the two coordinates. Make clear these are estimates from the iteration, not copied from the analytic answer in Q1.
- Discuss why the minimum value can appear more accurately determined than the point itself. Use the local contour shape, flat or shallow directions, and the size of recent decreases as evidence.
- Keep the prose focused on what the results show; do not describe the program beyond what is needed to explain the line searches.

Insert figure

`figures/q2_steepest_function4_path.pdf`

Iteration path from the specified starting point, with contour background if useful.

Figure 2: Steepest-descents path for function (4).

Estimate the limiting minimum value and coordinate intervals using only the numerical run. Then comment on why the value may be better determined than the location, linking this to the local shape of the function.

Question 3: Function (5)

What to report.

- Required evidence: a path plot for a chosen run of 9–15 iterations and a compact trace table with λ^*, values, and decreases. If the full table is too long, keep enough rows to show the pattern and state where omitted rows are represented.
- State the point to which the method appears to converge and support this using the iteration path and recent coordinates, without relying solely on the known analytic answer.
- Comment on the convergence rate and sensitivity to small changes in the line-search choices. A short comparison of two nearby λ^* choices or paths is enough if it shows the effect clearly.

Table 3: Steepest-descents trace for function (4).

n	x_n	y_n	λ_n^*	$f(x_n)$	Decrease
0	$\cdots$	$\cdots$	–	$\cdots$	–
1	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$
2	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$
3	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$
4	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$
5	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$
6	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$
7	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$
8	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$
9	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$
10	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$

- Relate any inefficiency to the geometry of the objective, such as elongated or curved valleys and repeated changes of direction, rather than to coding details.

Insert figure

`figures/q3_steepest_function5_path.pdf`

Iteration path for 9–15 iterations; add contours if this helps show slow progress.

Figure 3: Steepest-descents path for function (5).

State the likely limiting point, comment on convergence rate, and summarise one or two sensitivity checks for small changes in λ^*. Explain the geometric circumstances under which steepest descents is inefficient.

Table 4: Steepest-descents trace for function (5).

n	x_n	y_n	λ_n^*	$f(x_n)$	Decrease
0	$\cdots$	$\cdots$	–	$\cdots$	–
1	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$
2	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$
3	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$
4	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$
5	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$
$\vdots$					
m	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$

Questions 4–5: Conjugate Gradients

Question 4: Function (4)

What to report.

- Use the same reporting format as Q2 so the comparison is immediate: path plot, trace table, final numerical estimates, and line-search values.
- Identify where conjugate-gradient steps differ from steepest descents, especially after the first step and after any restart. A "step type" column is enough; no long derivation of the algorithm is needed.
- Compare against Q2 using observed progress per iteration, path geometry, and final value/location estimates. The answer should say whether the method improves the Q2 run for this function, and in what sense.

Insert figure

`figures/q4_cg_function4_path.pdf`

Conjugate-gradient path for the same start and iteration count as Q2.

Figure 4: Conjugate-gradient path for function (4).

Compare with Q2 using the same quantities: path geometry, final value, estimated minimiser, and sensitivity to the chosen line searches.

Question 5: Function (5)

What to report.

- Mirror Q3: give the conjugate-gradient path, trace table, line-search values, and the same kind of convergence comments.

Table 5: Conjugate-gradient trace for function (4).

n	x_n	y_n	λ_n^*	$f(x_n)$	Decrease	Step type
0	$\cdots$	$\cdots$	–	$\cdots$	–	start
1	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$	SD
2	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$	CG/restart
$\vdots$						
10	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$

- Compare directly with Q3 on the quantities the assessor can see: number of iterations, decreases in f, shape of the path, and apparent sensitivity to λ^*.
- End with a short answer to whether conjugate gradients gives much improvement over steepest descents across Q4 and Q5. Keep the conclusion tied to the two tested functions, not to general textbook claims.

Insert figure

`figures/q5_cg_function5_path.pdf`

Conjugate-gradient path for the same start and comparable iteration count as Q3.

Figure 5: Conjugate-gradient path for function (5).

Table 6: Conjugate-gradient trace for function (5).

n	x_n	y_n	λ_n^*	$f(x_n)$	Decrease	Step type
0	$\cdots$	$\cdots$	–	$\cdots$	–	start
1	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$	SD
2	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$	CG/restart
$\vdots$						
m	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$

Compare with Q3, then answer whether conjugate gradients offers much improvement over steepest descents for these two functions. Keep this as a short evidence-based paragraph rather than a general essay.

Questions 6–8: DFP Algorithm

Question 6: Quadratic Function (6)

What to report.

- Required evidence: the three DFP steps from the given starting point using the given λ^* sequence, including the final point, final value, and printed H matrix.
- Check the special quadratic property by comparing the three-step result with the expected behaviour for a quadratic in three variables. Keep this numerical and concise.
- Investigate sensitivity by changing the supplied λ^* values slightly and reporting a small selection of outcomes. A table of perturbations is more concise than several paragraphs.
- Verify that H tends to the inverse Hessian by comparing entries or by reporting a matrix norm of the difference. Do not print every intermediate H unless it supports a specific point.

Table 7: DFP trace for the three-dimensional quadratic.

n	x_n	y_n	z_n	λ_n^*	$f(x_n)$
0	$\cdots$	$\cdots$	$\cdots$	–	$\cdots$
1	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$
2	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$
3	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$

$$H_3 = \begin{pmatrix} \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots \end{pmatrix}$$

Table 8: Sensitivity of the three-iteration DFP result to line-search perturbations.

Perturbation to λ^* values	Final point	Final value	Comment
none	$\cdots$	$\cdots$	$\cdots$
$\cdots$	$\cdots$	$\cdots$	$\cdots$
$\cdots$	$\cdots$	$\cdots$	$\cdots$

Verify that the reported H is tending to the inverse Hessian by comparing selected entries or a norm of the difference. Comment on sensitivity to line-search errors; do not rederive the DFP update.

Question 7: Function (4)

What to report.

- Repeat the Q2 evidence using DFP: path plot, trace table, λ^* values, and a brief estimate of the limiting point/value.
- Include the printed final or representative H matrix, then compare it with the true inverse Hessian at the relevant point. State exactly what point was used for the Hessian comparison.
- Compare with Q2 and Q4 in terms of convergence, path shape, and how much curvature information appears to help for function (4).

$$H_m = \begin{pmatrix} \cdots & \cdots \\ \cdots & \cdots \end{pmatrix}, \qquad G(\text{comparison point})^{-1} = \begin{pmatrix} \cdots & \cdots \\ \cdots & \cdots \end{pmatrix}.$$

Compare the final or representative H with the true inverse Hessian at the relevant point. Then compare the DFP run with Q2 and Q4.

Insert figure

`figures/q7_dfp_function4_path.pdf`

DFP path for function (4), comparable to Q2 and Q4.

Figure 6: DFP path for function (4).

Table 9: DFP trace for function (4).

n	x_n	y_n	λ_n^*	$f(x_n)$	Decrease
0	$\cdots$	$\cdots$	–	$\cdots$	–
1	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$
2	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$
$\vdots$					
m	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$

Question 8: Function (5)

What to report.

- Repeat the Q3 evidence using DFP: path plot, trace table, λ^* values, and comments on convergence toward the apparent limiting point.
- Include the printed final or representative H matrix and compare it with the true inverse Hessian at the chosen comparison point. Mention whether the entries are reliable enough to support the comparison.
- Compare with Q3 and Q5, especially for the narrow-valley behaviour and the amount of progress achieved per iteration.

Insert figure

`figures/q8_dfp_function5_path.pdf`

DFP path for function (5), comparable to Q3 and Q5.

Figure 7: DFP path for function (5).

Table 10: DFP trace for function (5).

n	x_n	y_n	λ_n^*	$f(x_n)$	Decrease
0	$\cdots$	$\cdots$	–	$\cdots$	–
1	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$
2	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$
$\vdots$					
m	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$

$$H_m = \begin{pmatrix} \cdots & \cdots \\ \cdots & \cdots \end{pmatrix}, \qquad G(\text{comparison point})^{-1} = \begin{pmatrix} \cdots & \cdots \\ \cdots & \cdots \end{pmatrix}.$$

Compare the final or representative H with the true inverse Hessian at the relevant point. Then compare the DFP run with Q3 and Q5.

Question 9: Overall Comparison

What to report.

- This is the synthesis question: compare all three methods using the numerical evidence already shown, not new bulk output.
- Use a compact table for final values/locations and iteration counts, then one concise paragraph on the main pattern for function (4) and one for function (5), or one combined paragraph if the conclusion is simple.
- Address accuracy, sensitivity to line searches, path shape, and whether curvature information improved the method. Avoid a general essay on optimisation methods.

Table 11: Performance comparison of the three methods.

Function	Method	Iterations used	Final value/location summary	Main observation
(4)	Steepest descents	$\cdots$	$\cdots$	$\cdots$
(4)	Conjugate gradients	$\cdots$	$\cdots$	$\cdots$
(4)	DFP	$\cdots$	$\cdots$	$\cdots$
(5)	Steepest descents	$\cdots$	$\cdots$	$\cdots$
(5)	Conjugate gradients	$\cdots$	$\cdots$	$\cdots$
(5)	DFP	$\cdots$	$\cdots$	$\cdots$

Use this final paragraph to answer the comparison question directly. Discuss accuracy, iteration-path shape, sensitivity to λ^*, and whether each method exploited curvature information. Keep the conclusion tied to the two functions actually tested.

References

References

[1] R. Fletcher and M. J. D. Powell, *Rapidly convergent descent method for minimisation*, Computer Journal, 7 (1963).